

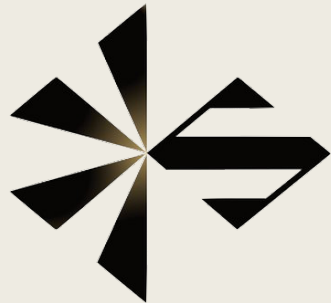


Non-Traditional Machining Processes – Part 1

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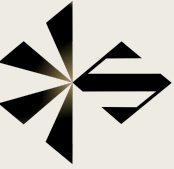
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Non-Traditional Machining Processes

Content

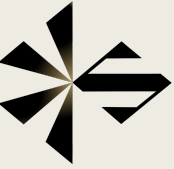
I) Conventional Machining vs Non-Traditional machining

II) Chemical and Electrochemical Processes

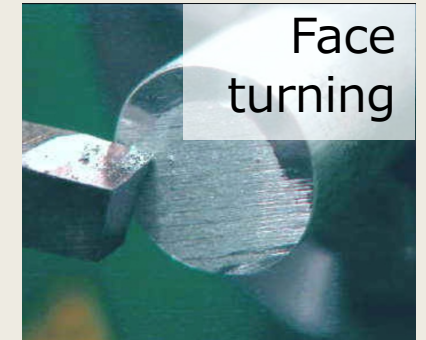
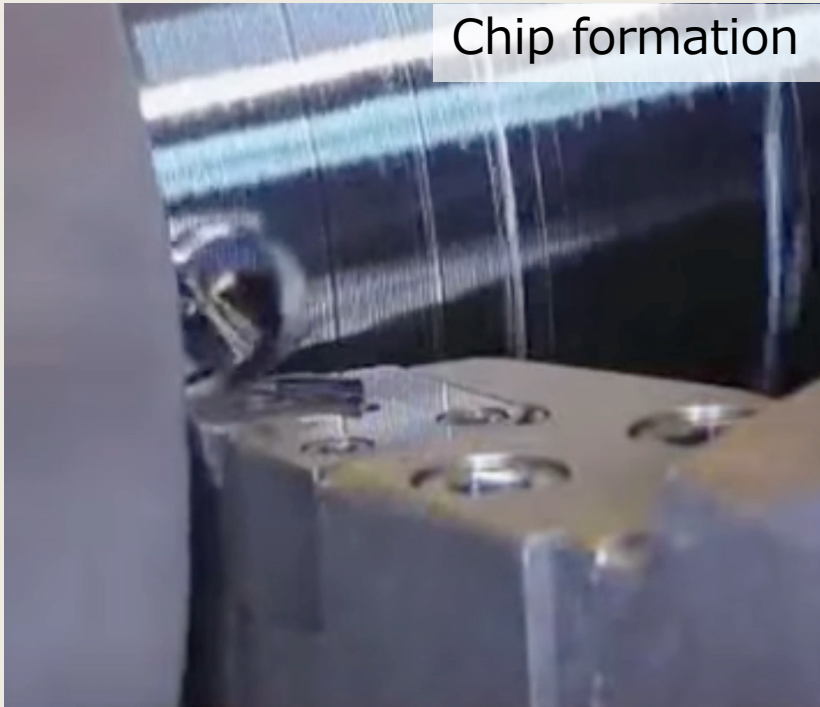
III) Mechanical Processes

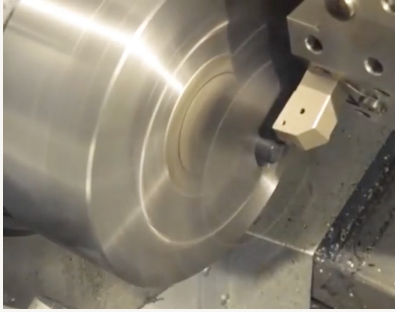
IV) Thermal Processes

V) Synthesis and conclusion

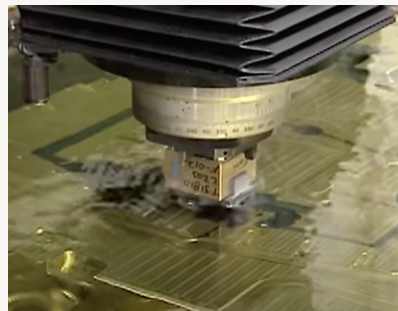


Conventional Machining Processes





I) Conventional Machining vs Non-Traditional Machining Processes

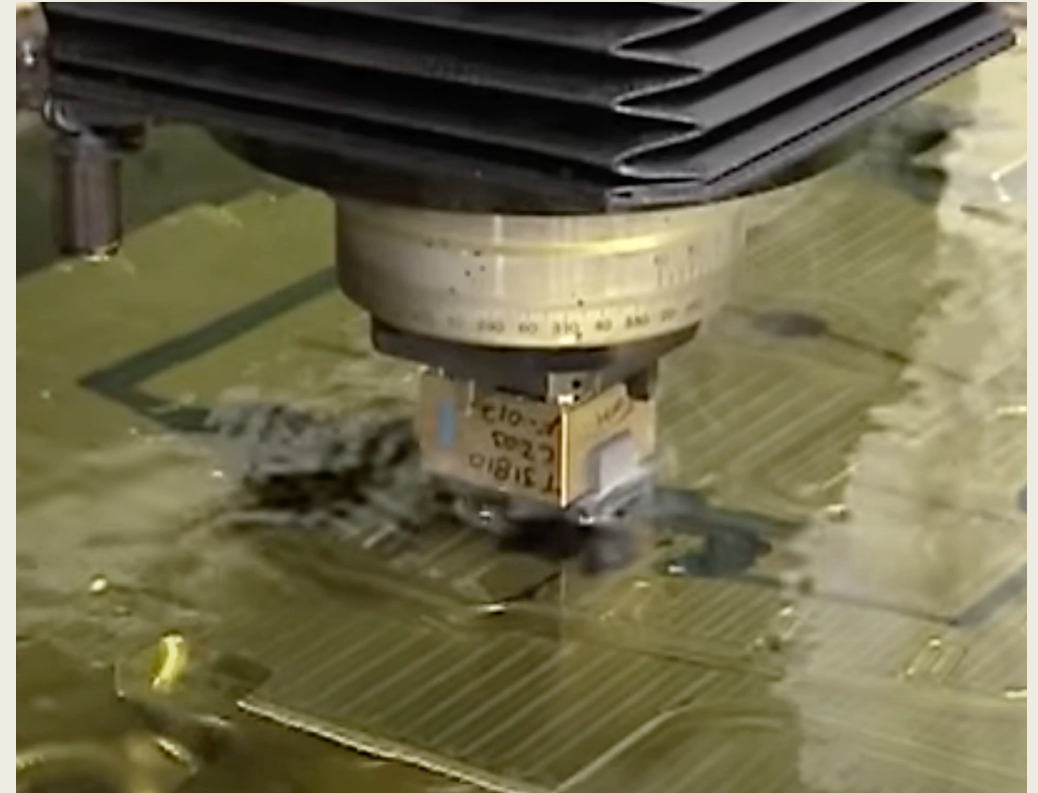


Conventional Machining vs Non-Traditional Machining Processes



Similarities

- Both are subtractive processes.
- Both can produce sufficiently smooth and finished surface with close tolerance depending on the process capability.





Conventional Machining vs Non-Traditional Machining Processes

Differences

Conventional Machining	Non-Traditional Machining (NTM)
Only mechanical energy is used for material removal	Various forms of energy are used to remove excess material.
Chip formation	No chips, but tiny metal particles, ions, molten or vapor, etc. depending on the process
Plastic deformation till fracture mechanisms cause material removal	Material mechanical behavior plays an insignificant role in material removal (examples: erosion, dissolution, evaporation, sputtering)
Cutting tool having specific geometry, material and properties is required	In few cases a tool is required, and its profile and property must match with the requirement.
Physical contact between the cutting tool and the workpiece	In few processes, solid abrasive grits strike the work surface to erode material.



Conventional Machining vs Non-Traditional Machining Processes

Differences

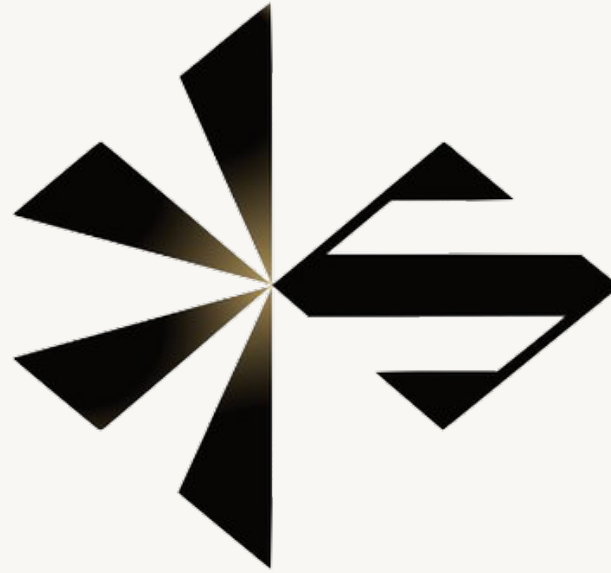
Conventional Machining	Non-Traditional Machining (NTM)
A well mastered process, cheaper costs and qualified operators easy to find.	Developing processes, expensive and qualified operators cost is high.
A wide range of surface finish depending on the process. Superfinishing quality ($<10\mu\text{m}$) requires costly CNC systems.	Surface finish is much better than that for conventional processes. Superfinishing quality in nanometer range.
Suitable for bulk removal of material, fast, reliable, and economic providing higher material removal rate (MRR)	Suitable for highly accurate and precise requirements. Time consuming, less reliable, and uneconomic for bulk material removal.
Operational on a limited number of shapes	Suitable for producing critical shapes

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